Plane mirrors

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Learning target: Use the reflection law and locate a virtual image.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 10.*
Correct speed=2×50/0.5=200 m/s. The student used half the traveled path and gets half the speed.

02. Worked example: explain the model
Image behind mirror at equal perpendicular distance; backward extensions are not actual rays behind an opaque mirror.

03. Read and interpret the model
Reflected ray 30° on other side of normal.

04. Apply the physics
2 m behind; 4 m apart.

05. Explain your reasoning
Reflected rays appear to originate behind mirror; they lack actual convergence there.

06. Review: connect an earlier idea *Review: Unit 6, Day 1.*
Need wave speed; target distance=speed×4/2. The pulse travels to the target and back.

07. Exit check
45°.

08. STRETCH: test the model
Image 1.5 m behind; separation 3 m.

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Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 10.*
True speed 340 m/s; incorrect speed≈291.4 m/s, biased low.

02. Worked example: explain the model
Image behind mirror at equal perpendicular distance; backward extensions are not actual rays behind an opaque mirror.

03. Read and interpret the model
Reflected ray 30° on other side of normal.

04. Apply the physics
70° from normal; 140° between the rays. Angle based on rays drawn away from point of incidence.

05. Explain your reasoning
Reflected rays appear to originate behind mirror; they lack actual convergence there.

06. Review: connect an earlier idea *Review: Unit 6, Day 1.*
Second/first range=6/4≈1.5. Delay does not establish mass.

07. Exit check
Image moves 0.50 m toward mirror from behind; separation decreases 1.0 m.

08. STRETCH: test the model
Image moves toward mirror from behind at 1.2 m/s; separation decreases 2.4 m/s.